

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Research Assessment

**COURSE NAME: ARTIFICIAL INTELLIGENCE
AND EXPERT SYSTEMS**

COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 45 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 5

Annexure A

Scenario-Based Research Assessment

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion Weightage Marks Obtained

Criterion	Weightage	Marks Obtained
Identification of Latest AI Application	10%	
Problem Analysis and Domain Understanding	15%	
AI Technologies and Working Mechanism	20%	
Research Quality and Literature Review	10%	
Real-World Implementation and Case Analysis	10%	
Impact, Challenges, and Future Scope	15%	
Originality and Critical Thinking	10%	
Organization, Presentation, and Documentation	10%	
Total	100	

Signature of the Faculty

Total Marks Awarded

AI Domain Scenario-Based Research Report

1. Title of AI Application

Item	Details
Title	AI-Assisted Medical Diagnosis Using Generative Artificial Intelligence
AI Domain	Healthcare

2. Domain Selection

Domain	Description
Healthcare	Artificial Intelligence is transforming healthcare by improving disease diagnosis, medical imaging, patient monitoring, treatment planning, and clinical decision-making. AI helps doctors analyze large amounts of medical data quickly and accurately, resulting in better patient care and reduced diagnostic errors.

3. Introduction and Background

Artificial Intelligence (AI) is changing the healthcare industry by providing intelligent solutions for disease diagnosis and treatment. Modern AI systems can analyze patient symptoms, medical records, laboratory reports, and medical images within a few seconds. Healthcare organizations use AI to improve diagnostic accuracy, reduce workload, and support doctors in making faster clinical decisions.

Recent advancements in Generative AI and Large Language Models (LLMs) have further improved healthcare by generating clinical summaries, answering medical queries, and assisting physicians during diagnosis. AI acts as a decision-support tool that enhances healthcare services without replacing medical professionals.

4. Problem Identification

Healthcare professionals face several challenges while diagnosing diseases.

Problems	Description
Large Patient Volume	Hospitals receive thousands of patients every day.
Human Errors	Fatigue and workload may affect diagnosis accuracy.

Problems	Description
Delayed Diagnosis	Manual analysis requires more time.
Limited Specialists	Rural areas often lack experienced doctors.
Huge Medical Data	Electronic records and medical images are difficult to analyze manually.

These challenges create the need for intelligent healthcare systems.

5. Need for AI Solution

Traditional diagnosis depends mainly on human experience. AI improves healthcare by processing large medical datasets faster and identifying hidden disease patterns.

Benefits of AI over Traditional Methods

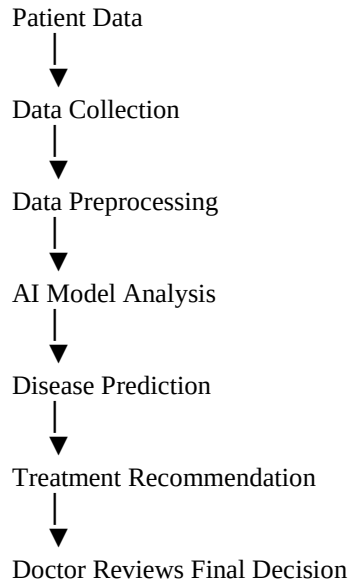
Traditional Method	AI-Based Method
Manual diagnosis	Automated analysis
Time-consuming	Fast diagnosis
Higher risk of errors	Improved accuracy
Limited data analysis	Big data processing
Doctor-dependent	AI-assisted decision support

6. AI Technologies Used

AI Technology	Application
Machine Learning	Disease prediction
Deep Learning	Medical image analysis
Natural Language Processing	Clinical note analysis
Computer Vision	X-ray, MRI, CT scan analysis
Generative AI	Clinical report generation
Large Language Models (LLMs)	Medical question answering
Predictive Analytics	Disease risk prediction

7. Working Mechanism

The AI-assisted medical diagnosis system follows these steps:



Workflow Description

Step Activity

- 1 Collect patient information
- 2 Clean and preprocess data
- 3 Analyze using AI models
- 4 Predict possible disease
- 5 Generate treatment suggestions
- 6 Doctor confirms diagnosis

8. Data Sources and Processing

Data Sources

Source	Examples
Electronic Health Records	Medical history
Medical Images	X-rays, MRI, CT scans
Laboratory Reports	Blood tests
Clinical Notes	Doctor observations
Wearable Devices	Heart rate, BP
Genetic Data	DNA information

Data Processing Steps

- Data Cleaning
- Data Integration
- Feature Extraction
- Model Training
- Disease Prediction

9. Real-World Implementation

Several healthcare organizations are successfully using AI-assisted medical diagnosis.

Organization	AI Application
Google Health	Cancer detection
Microsoft Healthcare AI	Clinical decision support
Mayo Clinic	Disease prediction
PathAI	Digital pathology
Tempus AI	Precision medicine
NVIDIA Healthcare	AI medical imaging

These organizations demonstrate how AI is improving healthcare quality worldwide.

10. Impact and Benefits

Benefit	Description
Faster Diagnosis	Quick analysis of medical data
Improved Accuracy	Reduced diagnostic errors
Early Disease Detection	Better treatment success
Reduced Costs	Lower healthcare expenses
Personalized Treatment	Patient-specific recommendations
Better Patient Care	Improved healthcare services
Increased Efficiency	Reduced doctor workload

Overall, AI improves healthcare quality while supporting medical professionals in delivering faster and more accurate treatment.

11. Challenges and Limitations

Although AI offers many advantages, several challenges still exist.

Challenge	Description
Poor Data Quality	Incomplete medical records
Privacy Issues	Protection of patient information
Algorithm Bias	Unfair predictions
High Cost	Expensive implementation
Lack of Explainability	Difficult to understand AI decisions
Ethical Concerns	Responsibility for AI decisions

Healthcare organizations must address these issues before large-scale AI adoption.

12. Future Scope and Emerging Trends

Future developments in AI-assisted healthcare include:

- Advanced Generative AI for clinical decision support
- Explainable AI (XAI)
- Precision Medicine
- AI-powered robotic surgery
- Wearable health monitoring
- Remote healthcare using AI
- Digital twins for personalized treatment
- Federated Learning for secure medical data sharing

These innovations are expected to improve healthcare quality and accessibility in the coming years.

13. Conclusion

Artificial Intelligence has become an essential technology in modern healthcare. AI-assisted medical diagnosis helps doctors analyze medical data more efficiently, detect diseases earlier, and provide personalized treatment recommendations. Technologies such as Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, and Generative AI have significantly improved diagnostic accuracy and healthcare efficiency.

Despite challenges related to privacy, security, ethics, and implementation costs, AI continues to evolve rapidly. With continuous research and responsible deployment, AI-assisted medical diagnosis will play an increasingly important role in improving patient care and building smarter healthcare systems.

14. References

Reference

- 1 World Health Organization (WHO), *Ethics and Governance of AI for Health*, 2021.
- 2 Esteva J. et al., "A Guide to Deep Learning in Healthcare," *Nature Medicine*, 2019.
- 3 Topol E., *Deep Medicine*, Basic Books, 2019.
- 4 Google Health – Artificial Intelligence in Healthcare.
- 5 Microsoft AI for Healthcare.
- 6 Mayo Clinic – Artificial Intelligence in Medicine.
- 7 *Nature Medicine Journal*.
IEEE Transactions on Medical Imaging.

Rubrics for Evaluation

S.No	Criteria	Weightage (%)	Excellent (100–90%)	Good (89–75%)	Satisfactory (74–60%)	Improvement ()
1.	Identification of Latest AI Application	10%	Selects a highly relevant, emerging AI application with strong justification.	Selects a recent and relevant AI application.	Selects a suitable application with limited justification.	Application is outdated or irrelevant.
2.	Problem Analysis and Domain Understanding	15%	Clearly explains the real-world problem, domain context, and need for AI with critical analysis.	Good explanation with adequate analysis.	Basic explanation with limited analysis.	Problem statement is unclear or incomplete.

3.	AI Technologies and Working Mechanism	20%	Accurately explains AI techniques, algorithms, architecture, and workflow with technical depth.	Good explanation with minor omissions.	Covers only basic AI concepts.	AI technologies are poorly explained or incorrect.
4.	Research Quality and Literature Review	10%	Uses recent, credible sources with proper citations and excellent synthesis.	Good quality references with appropriate citations.	Limited references with basic discussion.	Few or unreliable references; poor citation.
5.	Real-World Implementation and Case Analysis	10%	Thoroughly analyzes industrial implementation with appropriate examples.	Good implementation analysis.	Basic implementation discussion.	Minimal or no real-world analysis.
6.	Impact, Challenges, and Future Scope	15%	Critically evaluates benefits, limitations, ethical issues, and future trends.	Good evaluation with minor gaps.	Basic discussion of impact and future scope.	Limited evaluation or missing sections.
7.	Originality and Critical Thinking	10%	Demonstrates independent research, innovation, and critical evaluation.	Good analytical thinking.	Some originality with limited analysis.	Mostly descriptive or copied content.
8.	Organization, Presentation, and Documentation	10%	Exceptionally organized, professional presentation, error-free, and properly referenced.	Well organized with minor issues.	Adequately organized with some errors.	Poor organization and documentation.

